

INVERNESS AP, United Kingdom

WMO: 030590

Lat: 57.543N

Lon: 4.048W

Elev: 31

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.9	24.5	19.0	14.3	22.1	21.5	16.2	25.2	36.7	43.8	32.0	43.7	3.3	230	0.670

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	72.0	62.5	69.6	61.4	66.5	59.4	63.9	70.2	62.1	67.5	60.4	65.3	11.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	80.5	66.8	59.3	75.5	65.0	57.6	70.9	62.8	29.3	70.5	27.9	68.2	26.7	65.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.2	24.8	21.8	16.6	78.7	3.9	3.7	13.8	81.3	11.5	83.5	9.3	85.6	6.4	88.3		
			16.1	67.1	3.9	2.3	13.3	68.8	11.0	70.1	8.8	71.4	5.9	73.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	37.5	38.7	40.9	45.3	49.3	54.6	58.5	57.0	53.8	48.1	41.3	38.0	
	DBStd	9.02	5.70	5.64	5.40	5.20	4.87	4.34	4.22	3.58	4.34	5.09	5.43	7.13	
	HDD50	1974	388	317	284	159	71	8	0	1	13	95	263	375	
	HDD65	6586	853	736	746	592	486	311	207	249	336	522	710	838	
	CDD50	868	0	1	3	17	49	147	264	218	127	37	3	2	
	CDD65	6	0	0	0	0	0	0	5	1	0	0	0	0	
	CDH74	53	0	0	0	0	1	8	42	2	0	0	0	0	
	CDH80	4	0	0	0	0	0	1	3	0	0	0	0	0	
Wind	WSAvg	9.3	10.6	9.9	10.8	9.5	8.9	8.2	8.0	8.3	9.1	8.9	9.6	10.0	
Precipitation	PrecAvg	34.8	3.5	3.2	3.0	2.4	2.3	2.6	2.0	2.3	3.1	3.6	3.4	3.2	
	PrecMax	45.6	6.9	6.5	6.5	4.3	4.7	5.2	3.7	4.8	9.7	6.0	4.9	5.7	
	PrecMin	26.3	0.5	1.0	1.3	1.3	1.1	1.0	0.9	0.8	1.3	0.9	0.7	0.4	
	PrecStd	4.9	1.9	1.7	1.5	0.8	1.0	1.3	0.8	1.1	1.9	1.3	1.1	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	55.7	59.3	66.3	71.4	73.8	78.7	73.2	70.2	64.2	57.4	55.6	
		MCWB	49.8	50.4	50.5	54.7	60.2	61.6	64.6	63.0	61.9	57.7	53.6	51.9	
	2%	DB	50.3	53.2	53.9	61.0	66.3	69.4	73.7	69.5	67.0	61.1	55.0	53.6	
		MCWB	47.5	48.9	48.5	52.7	57.0	61.3	63.7	61.6	60.0	56.1	51.3	50.5	
	5%	DB	48.4	50.2	51.9	57.6	62.5	65.9	70.1	66.5	64.4	58.9	52.2	51.6	
		MCWB	46.1	46.7	47.0	50.8	55.4	58.6	62.1	59.6	58.1	54.4	48.6	48.5	
	10%	DB	46.5	48.0	50.0	55.4	59.1	62.8	67.6	64.6	62.3	57.0	50.3	48.6	
		MCWB	44.2	45.2	45.8	49.2	52.8	56.9	60.6	58.3	57.0	53.0	47.7	46.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	51.7	51.9	56.6	61.9	64.5	67.3	64.4	63.0	58.7	54.3	52.8
			MCDB	52.5	55.2	57.7	63.4	68.4	71.7	74.4	70.9	68.8	62.6	57.1	55.4
2%		WB	48.2	49.3	49.6	53.5	58.7	61.2	65.1	62.3	60.6	56.7	51.7	50.9	
		MCDB	50.3	52.5	53.9	59.4	64.4	67.1	71.8	67.2	65.5	60.6	54.4	53.3	
5%		WB	46.1	47.1	47.8	51.5	55.8	59.2	63.1	60.6	59.0	55.0	49.7	48.6	
		MCDB	48.3	50.3	51.7	57.3	60.9	64.6	69.0	65.4	63.6	58.3	51.8	51.0	
10%		WB	44.1	44.8	46.0	49.8	53.6	57.3	61.2	59.2	57.3	53.1	48.0	46.2	
		MCDB	46.0	47.1	49.4	54.5	58.4	62.4	66.0	63.7	61.1	56.5	50.4	48.7	
Mean Daily Temperature Range		5% DB	MDBR	10.1	12.5	14.0	16.2	16.6	14.9	15.6	14.9	14.7	14.0	11.4	10.9
			MCDBR	10.5	13.6	16.4	21.0	22.7	19.7	21.0	18.0	18.2	16.3	13.7	12.3
	MCWBR		9.3	10.7	12.0	14.0	14.9	12.5	12.3	11.8	12.5	12.4	11.7	11.1	
	5% WB	MCDBR	10.3	12.4	15.6	20.1	19.9	18.5	19.0	16.8	17.1	15.1	13.7	12.5	
		MCWBR	9.3	10.2	12.0	13.7	13.9	12.3	12.0	11.8	12.5	12.0	12.2	11.5	
Clear-Sky Solar Irradiance	taub	0.315	0.337	0.361	0.381	0.383	0.379	0.381	0.367	0.353	0.339	0.315	0.302		
	taud	2.214	2.296	2.295	2.283	2.326	2.371	2.375	2.421	2.458	2.412	2.271	2.154		
	Ebn at Noon	167	221	248	263	270	273	269	266	252	220	167	132		
	Edh at Noon	16	23	30	36	36	36	35	31	26	21	15	13		
All-Sky Solar Radiation	RadAvg	127	323	647	1074	1447	1415	1353	1071	750	395	168	88		
	RadStd	18	36	63	92	96	131	126	69	63	39	15	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page